Lecture 8 // Sept 17 // Singularities in $\mathcal{M}$ & Holonomy

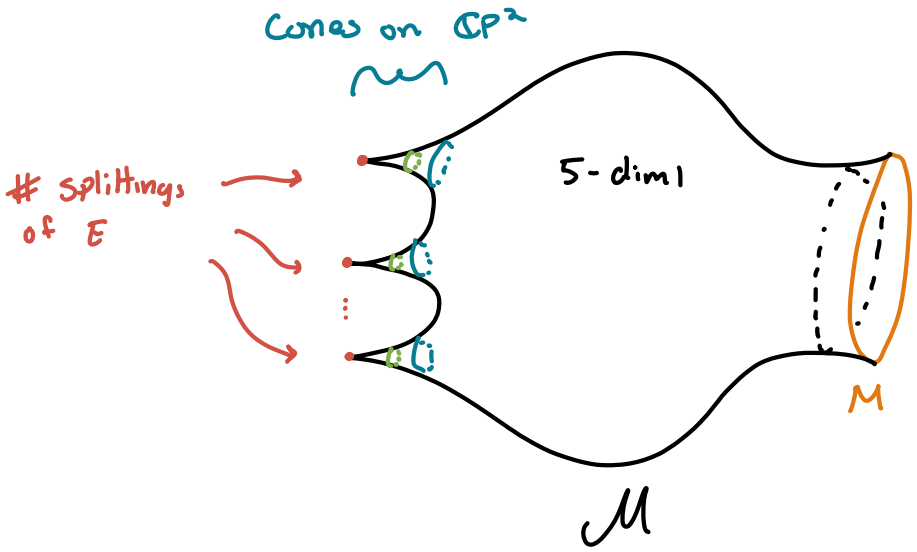
Last Time: Let (M^4, g) be positive definite and simply connected.

Fix a $k=1$ $SU(2)$ bundle E over M.

Consider $\mathcal{M} = \mathcal{A}/\mathcal{G}$ the SDYM moduli space.

We know the following qualitative features of $\mathcal{M}$:

1) Away from a finite number of singular points, $\mathcal{M}$ is a smooth 5-dim oriented manifold.
2) The number of these points = the # of splittings of E
3) Each point has a nbhd $\circ$ homeo to cone on $\mathbb{CP}^2$. Diffeo away from pt
4) $\mathcal{M}$ is <u>not</u> compact, but it admits a compactification $\overline{\mathcal{M}}$ such that $\partial\overline{\mathcal{M}} \cong M$



After some linear algebra, these properties prove Donaldson's theorem:

$$\text{rk } Q_M = \sigma(M) \overset{\text{Since } \sigma \text{ bordism inv.}}{=} \#\text{ Splittings of } E = \frac{1}{2}\#\, x \in H_2 \text{ with } x^2 = 1$$

($\sigma(M)$: Since definite)

Today, we'll unpack these points: why is $\mathcal{M}$ singular, and why do nbhds look like cone on $\mathbb{CP}^2$

Singularities in $\mathcal{M}$

Recall that $\mathcal{M} = \{\text{SDYM conn.}\}/\mathcal{G}$. $\{\text{SDYM CONN}\} \subseteq \mathcal{A}$, which is a smooth manifold, on which $\mathcal{G}$ acts.

However, $\mathcal{G}$ may have stabilizer subgroups.

Thus the singularities arise from the failure of $\mathcal{G}$ to act freely on $\mathcal{A}$.

Rmk Establishing $\{\text{SDYM conn}\}$ sm mfd is fairly standard:
It is the zero locus of map from oo-dim map to somewhere else,
verify that 0 is regular value, apply Sard-Smale.

Def For $\nabla \in \mathcal{A}$, $\text{Stab}(\nabla) = \{g \in \mathcal{G} \mid g^*(\nabla) = \nabla$

$$A \longrightarrow g^{-1}Ag + g^{-1}dg$$

What can $\text{stab}(\nabla)$ be? First note that for every ∇, $\pm Id \in \text{Stab}(\nabla)$.
Since $\pm Id$ is sent to itself under conjugation, $\pm Id$ is a well-defined section of $\text{Aut}(E)$ These are not interesting gauge transformations, so it's sometimes more convenient to take $\mathcal{G}/\pm Id \curvearrowright \mathcal{A}$ instead.

What else is possible? Let's examine an example.
connections with non trivial stabilizers are related to # splittings.
For a line bundle L, consider the constant gauge transformation given by multiplication by a fixed element of S^1, $e^{i\theta} \in S^1$.
For $E = L \oplus \bar{L}$, consider now $g = \begin{bmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{bmatrix}$

One can check this is a well defined gauge transformation on E.
If ∇ is split, eg, $A = \begin{bmatrix} ia & 0 \\ 0 & -ia \end{bmatrix}$, easy to check that $g^*(\nabla) = \nabla$
So if E is split and ∇ is a split connection, considered as a subgroup of $SU(2)$, $\text{Stab}(\nabla)$ is at least as big as S^1.

Rmk In this observation, we play fast and loose with the difference between a local definition of a gauge transformation and its definition globally. The correct statement is given in the following lemma, and the tension between global & local will be resolved as we introduce the tools to prove it.

Lemma For a $SU(2)$ bundle with $\kappa \neq 0$, a connection has stabilizer bigger than $\pm Id$ IFF it is split, and $Stab(D) \cong S^1$

The proof isn't hard, but requires more technical language.

But this lemma gives why the singularities of $\mathcal{M}$ are what they are. We showed previously that each splitting of E supports a unique (up to $\mathcal{G}$) split SDYM connections.

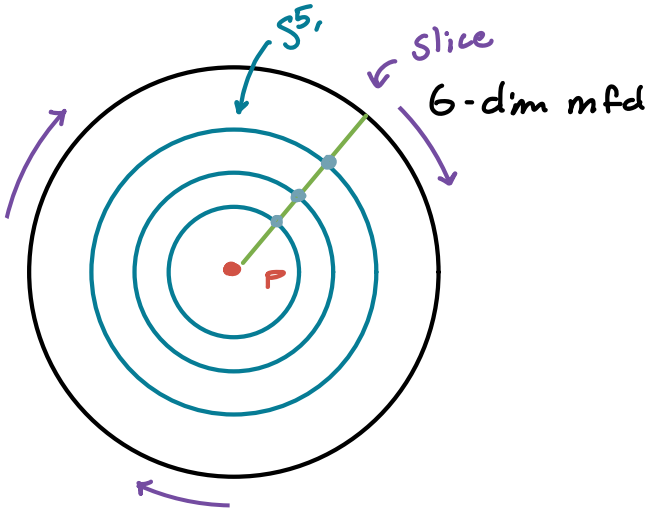
Each singular point is the unique SDYM connection for a splitting. Thus the # of singular points is the # of splittings. This gives 2)

We now ask why local neighborhoods are cones on $\mathbb{CP}^2$.

Near a singular point, $\mathcal{G}/\pm I$ acts freely.

The approximate idea is that we first quotient by $\pm I$ to get a 6-dim mfd with a $S^1/\pm I$ action which acts freely except at one point.

The local model for this is $S^1 \curvearrowright \mathbb{R}^6 \cong \mathbb{C}^3$ via usual S^1 complex multiplication. $\mathbb{C}^3/S^1 \cong$ cone on $\mathbb{CP}^2$ with one singular point, since on each S^5 slice, we have $S^5/S^1 \cong \mathbb{CP}^2$ (via definition $\mathbb{CP}^2 := S^{2n+1}/S^1$)



Note The $\pm I \subset \mathcal{G}$ and S^1 actions are more entertwined in $\mathcal{G}$ than this; we can't actually quotient by gauge in this "partial" way. But it gives an approximate idea.

Q Why is $\mathcal{M}$ 5-dimensional?

This is where we use that $k=1$.

The dimension is from the Atiyah Singer Index formula applied to a fredholm operator whose zero locus are the SDYM connections.

When $k>1$, we get more generally that Sing nbhd $\cong$ cone on $\mathbb{CP}^n$

We now must introduce more differential geometry.

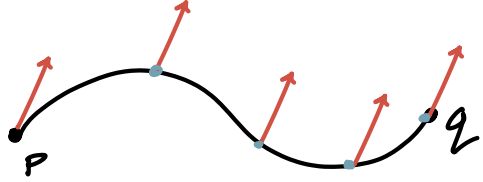
Interlude: Parallel Transport

Let E be a vector bundle over M.

Def Let ∇ be a connection on E, $\gamma : [0,1] \to M$ a path.
A section s defined along γ is parallel transported by ∇ along γ if $\nabla_{\gamma'(t)} s = 0$ for all t.

Lemma Let $\gamma : [0,1] \to M$ be a path from p to q.
If we fix any $v \in E_p$, then there exists a unique section s defined along γ such that s extends v and s is parallel transported by ∇ along γ.

Pf idea



Using the existance of solutions to ODE's, we can construct s and show that $(d+A)s = 0$ □

To better see the point of parallel transport, we can observe the effect of parallel transport on a frame of E_p, ie, a basis for E_p, and parallel transport that basis along γ.

Parallel transport preserves orthonormality, so this gives you a a particular trivialization of E along γ for which the basis elements are covariently constant: A vanishes in the path direction γ'.

In essence, the covariant derivative specifies a "preferred frame" along each path, which we can find by choosing a frame at one end and parallel transporting it along the path.

What else can parallel transport do?

For each path γ from p to q, we can build a linear map from E_p to E_q which sends $v \in E_p$ to the endpoint of the parallel transport of v along γ. Call this map $hol_{p,q}(\gamma): E_p \to E_q$. Since $hol_{p,q}(\gamma) \circ hol_{p,q}(\gamma) = id$, this map is an isomorphism.

Now, take $q = p$, so γ is is a based loop in M.

Then $hol_p = hol_{p,p}$ gives a map $E_p \to E_p$, so defines an element $hol_p(\gamma) \in GL(E_p)$.

Def Holonomy is a map $\text{hol}_p : \{ \text{loops based at } p \} \to GL(E_p)$

Properties of Holonomy:

1) If E is a G-bundle and ∇ is a G connection, then $\text{hol}_p(\gamma)$ is $G \leq GL(E_p)$ valued

2) If $F_\nabla = 0$, the holonomy only depends on the based smooth homotopy class of γ.

   Ie, hol_p descends to a map

   $$\text{hol}_p : \pi_1(M) \to GL(E_p) \quad (\text{or } \mathfrak{g} \text{ for } G)$$

   For $G = SU(2)$, $\text{hol}_p : \pi_1(M) \to su(2)$ is a $su(2)$ representation of $\pi_1(M)$.

3) If g is a gauge transformation,

   $$\text{hol}^{g^*(\nabla)}_{p,q}(\gamma) = g_q^{-1} \circ \text{hol}^{\nabla}_{p,q}(\gamma) \circ g_p$$

To illustrate holonomy, consider a trivial line bundle $L = M \times \mathbb{C}$ over M and a $U(1)$ connection on L given by $\nabla = d + ia$, $a \in \Omega^1(M; i\mathbb{R})$.

Thus a section here is just a $\mathbb{C}$-valued function. For a path γ, what does it mean for $\nabla_{\gamma'(t)} s = 0$? Given a based loop γ and a section defined along γ, what does it mean for s to be parallel transported along γ? Since we're working in a line bundle, a section is just a $\mathbb{C}$-valued function, and for $\nabla_{\gamma'(t)} s = 0$, the covariant derivative of this function must vanish along γ.

Explicitly, we get

$$\nabla_{\gamma'(t)} s = 0$$

$$(ds)(\gamma'(t)) + ia(\gamma'(t))s = 0$$

$$\frac{d}{dt}(s \circ \gamma) + ia(\gamma'(t))(s \circ \gamma(t)) = 0$$

But this is just a differential equation, which we can solve as usual:

$$\frac{d}{dt}(s\circ\gamma) = -ia(\gamma'(t))(s\circ\gamma(t))$$

$$\Rightarrow (s\circ\gamma) = \left(e^{-i\int_0^t a(\gamma'(h))dh}\right) s(\gamma(0))$$

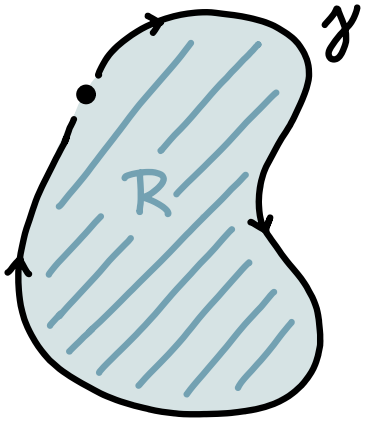
If we conflate s and $s\circ\gamma$, we get that $hol_{\gamma,\nabla} = e^{-i\int_\gamma a} \in S^1$.

Intuitively, we're supposed to think that solving $(d+A)s=0$ gives $s(t) = e^{-\int A} s(0)$, so if A is lie alg. valued, $-\int A$ lie alg. valued, so $e^{-\int A}$ lie gp valued. In reality, this is false – there's an issue with commutivity when $G \neq U(1)$.

However we see directly here that $hol_\nabla(\gamma)$ need not be the identity: if $\int_\gamma a \neq 2\pi n$, $e^{-i\int_\gamma a} \neq 1$.

In fact, we see a nice connection with property #3. Indeed, suppose that γ bounds a region R.

Then Stokes' theorem says that



$$\int_\gamma a = \int_{\partial R} a = \int_R da$$

where da is the curvature of ∇

So the heuristic is that holonomy picks up the curvature of a connection.

This demonstrates, in the $U(1)$ case, assertion #3:

If γ is varied through region where $F_\nabla = 0$, then holonomy is not changed.

However, note that this analysis applies to single coordinate charts; it is purely local. If the bundle is nontriv., it is not true that $F_A = 0$ implies that $\mathrm{hol}(\gamma) = id$. Furthermore, it makes no sense to treat the differential equation like its globally defined, since a isn't. But it is true that holonomy is const. in its homotopy class.

Next time: We'll use parallel transport to discuss the lemma on the exact behaviour of stabilizers of connections.